

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA05

COURSE OUTCOME COVERED

CO2: *Apply image enhancement and segmentation techniques for extracting meaningful regions from images. (BTL 3)*

Assessment Tool Weightage: 50%

1. Grey Level Transformation

- (a) Explain the purpose of grey level transformation in image enhancement.
- (b) Apply a negative transformation $s = 255 - r$ to the pixel values:

$$r = [50, 100, 150, 200]$$

Compute the transformed values.

2. Log Transformation

- (a) Explain how log transformation enhances low-intensity pixels.
- (b) Given $s = c \log(1 + r)$, where $c = 10$, compute s for:

$$r = [0, 10, 50, 100]$$

3. Histogram Processing

- (a) Explain the role of histogram equalization in contrast enhancement.
- (b) Given a 3-bit image with histogram:

Intensity Frequency	
0	2
1	2
2	4

Intensity Frequency

3 8

Compute the cumulative distribution and equalized intensity values.

4. Spatial Filtering (Smoothing)

(a) Explain the purpose of spatial filtering for smoothing.

(b) Apply a 3×3 averaging filter to the following region:

[10 20 30 20 30 40 30 40 50]

Compute the new center pixel value.

5. Spatial Filtering (Sharpening)

(a) Explain how sharpening enhances image details.

(b) Apply the Laplacian mask:

$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix}$

to the matrix:

[10 10 10, 10 20 10, 10 10 10]

Compute the output at the center.

6. Frequency Domain Filtering

(a) Differentiate low-pass and high-pass filtering in frequency domain.

(b) Given frequency components:

$F = [100, 80, 40, 20]$

Apply a low-pass filter that retains only the first two components. Write the filtered output.

7. Thresholding

(a) Explain global thresholding in segmentation.

(b) Given pixel values:

[200]

Apply threshold $T = 100$. Assign values (0 or 1).

8. Edge Detection

(a) Explain the concept of edge detection.

(b) Apply the Sobel operator (horizontal):

$\begin{bmatrix} -1 & -2 & -1, & 0 & 0 & 0, & 1 & 2 & 1 \end{bmatrix}$

to:

[10 10 10, 20 20 20, 30 30 30]

Compute the gradient at the center.

9. Region-Based Segmentation

(a) Explain region growing in segmentation.

(b) Given pixel intensities:

[1 0 2]

If seed = 50 and threshold difference = 5, identify pixels belonging to the region.

10. Combined Enhancement & Segmentation

(a) Explain why smoothing is applied before segmentation.

(b) Given image values:

[1 0 5]

First apply averaging (assume mean = 48), then threshold at $T = 50$.

Determine final segmented output.

11. Power-Law (Gamma) Transformation

A grayscale image is enhanced using power-law transformation.

(a) Interpret the role of gamma correction in image enhancement.

(b) Justify the use of transformation $s = cr^\gamma$, where $\gamma = 0.5$, $c = 1$.

(c) Apply the transformation step-by-step to pixel values:

$r = [25, 100, 225]$.

(d) Compute the transformed values accurately.

(e) Interpret the results in terms of brightness enhancement.

12. Contrast Stretching

An image has intensity values in the range [50, 150].

(a) Interpret the need for contrast stretching.

(b) Justify linear transformation to range [0, 255].

(c) Apply transformation step-by-step for values:

$r = [50, 100, 150]$.

(d) Compute the output values accurately.

(e) Interpret improvement in contrast.

13. Histogram Equalization (4-bit Image)

A 4-bit image has the following histogram:

Intensity Frequency

0	4
1	6
2	10
3	6

- (a) Interpret the need for equalization.
- (b) Justify use of CDF.
- (c) Compute cumulative distribution.
- (d) Compute new intensity levels.
- (e) Interpret contrast enhancement.

14. Median Filtering

A noisy image contains salt-and-pepper noise.

- (a) Interpret noise characteristics.
- (b) Justify median filter usage.
- (c) Apply filter to:

10 200 10,200 50 200,10 200 10

- (d) Compute center value.
- (e) Interpret noise removal effectiveness.

15. Gaussian Smoothing

An image is smoothed using Gaussian filter.

- (a) Interpret smoothing objective.
- (b) Justify Gaussian filter over averaging.
- (c) Apply filter:

[1 2 1, 2 4 2, 1 2 1]

to matrix:

[10 20 10,20 40 20,10 20 10]

- (d) Compute center pixel.
- (e) Interpret smoothing effect.

16. High-Pass Filtering

An image requires sharpening.

- (a) Interpret need for high-pass filtering.
- (b) Justify filter selection.

(c) Apply mask:

$[-1 \ -1 \ -1, -1 \ 8 \ -1, -1 \ -1 \ -1]$

(d) Compute center value.

(e) Interpret edge enhancement.

17. Frequency Domain High-Pass

Given frequency components:

$$F = [10, 20, 80, 100]$$

(a) Interpret high-frequency importance.

(b) Justify high-pass filtering.

(c) Apply filtering (remove low frequencies).

(d) Compute output.

(e) Interpret result.

18. Adaptive Thresholding

Pixel values: [60, 80, 120, 140]

(a) Interpret segmentation challenge.

(b) Justify adaptive thresholding.

(c) Apply local threshold (mean = 100).

(d) Compute binary output.

(e) Interpret result.

19. Edge Detection (Vertical Sobel)

Apply vertical Sobel:

$$[-1 \ 0 \ 1, -2 \ 0 \ 2, -1 \ 0 \ 1]$$

Matrix:

$$[10 \ 20 \ 30, 10 \ 20 \ 30, 10 \ 20 \ 30]$$

(a) Interpret edge detection objective.

(b) Justify operator choice.

(c) Apply step-by-step.

(d) Compute center value.

(e) Interpret edge direction.

20. Prewitt Edge Detection

(a) Interpret Prewitt operator purpose.

(b) Justify use over Sobel.

(c) Apply kernel:

$[-1 \ 0 \ 1, -1 \ 0 \ 1, -1 \ 0 \ 1]$

(d) Compute output.

(e) Interpret result.

21. Region Splitting

Pixel set: [40, 45, 100, 105]

(a) Interpret segmentation problem.

(b) Justify splitting method.

(c) Apply threshold 80.

(d) Compute regions.

(e) Interpret separation.

22. Region Merging

Regions: [45, 50], [52, 55]

(a) Interpret merging criteria.

(b) Justify merging approach.

(c) Apply threshold = 5.

(d) Compute merged region.

(e) Interpret result.

23. Edge Linking

Edges detected: [1, 0, 1, 0, 1]

(a) Interpret problem of discontinuity.

(b) Justify linking method.

(c) Apply linking step.

(d) Compute result.

(e) Interpret continuity.

24. Low-Pass Filtering (Frequency)

Given $F = [200, 150, 50, 20]$

(a) Interpret smoothing requirement.

(b) Justify LPF selection.

(c) Apply filter.

- (d) Compute output.
- (e) Interpret effect.

25. Laplacian Sharpening (Variant)

Mask:

$$\begin{array}{ccc} 0 & -1 & 0, \\ -1 & 5 & -1, \\ 0 & -1 & 0 \end{array}$$

- (a) Interpret sharpening need.
- (b) Justify mask.
- (c) Apply to matrix.
- (d) Compute center.
- (e) Interpret result.

26. Histogram Matching

- (a) Interpret need for matching.
- (b) Justify method.
- (c) Map values [10, 50, 100].
- (d) Compute new values.
- (e) Interpret.

27. Noise Reduction Comparison

- (a) Interpret noise type.
- (b) Compare averaging vs median.
- (c) Apply both.
- (d) Compute outputs.
- (e) Interpret differences.

28. Combined Filtering

- (a) Interpret need for smoothing + sharpening.
- (b) Justify sequence.
- (c) Apply filters stepwise.
- (d) Compute output.
- (e) Interpret improvement.

29. Multi-Threshold Segmentation

Pixels: [20, 60, 120, 180]

- (a) Interpret multi-level segmentation.
- (b) Justify thresholds (50,100).
- (c) Apply segmentation.
- (d) Compute classes.
- (e) Interpret.

30. Boundary Detection

- (a) Interpret boundary extraction.
- (b) Justify method.
- (c) Apply edge + linking.
- (d) Compute result.
- (e) Interpret boundaries.

Code of Conduct:

*I R. Sai Pavan /192472076 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: R. Sai Pavan

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

Analytical problem solving R. sat Pavan

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1924-2076

1) Grey level Transformation:

(a) Apply a negative transformation $s = 255 - r$ to the pixel values: $r = [50, 100, 150, 200]$

(b) Apply the purpose of grey level transformation in image enhancement.

compute the transformed values.

Answer:

a) * Improve image brightness and contrast
* enhance image quality

b) Formula: $s = 255 - r$

r	s
50	205
100	155
150	105
200	55

2) log Transformation:

(a) explain how log transformation enhances low intensity pixels.

(b) Given $s = c \log(1+r)$, where $c > 10$, compute s for $r = [0, 10, 50, 100]$.

Answers:

a) * enhances dark pixels
* compresses bright pixels.

b) formula $s = 10 \log(1+r)$

r	s
0	0
10	10.4
50	17.18
100	20.24

3. Histogram processing:

- Explain the role of histogram equalization in contrast.
- Given a 3 bit image with histogram

Intensity	frequency
0	2
1	2
2	4
3	8

compute the cumulative distribution and equalized intensity values.

- * Improves image contrast
* Distributes pixel intensities uniformly.

b) CDF:-

Intensity	CDF	New Intensity
0	2	1
1	4	2
2	8	4
3	16	7

4. Spatial Filtering

- Explain the purpose of spatial filtering for smoothing.
- Apply 3×3 averaging filter to the region $\begin{bmatrix} 10 & 20 & 30 & 20 & 30 & 40 & 50 \end{bmatrix}$
compute the new center pixel value

(a) Removes noise

* produces a smooth image

$$(b) \frac{10 + 20 + 30 + 20 + 30 + 40 + 50}{9} = \frac{270}{9} = 30$$

Spatial filtering (Sharpening)

(a) Explain how sharpening enhances image details.

(b) Apply the Laplacian mask $\begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix}$

$$\begin{bmatrix} 10 & 10 & 10 & 10 & 20 & 10 & 10 & 10 \end{bmatrix}$$

(a) * Enhanced edges

* Improved image details

(b) $0(-10) - 10 + 0 - 10 + 4(20) - 10 + 0 - 10 + 0$
 $= 40.$

6) (a) * Low pass filter removes high-frequency components
* High-pass filter removes low-frequency components

(b) $F = [100, 80, 40, 20]$

Retain first two components

$$[100, 80, 0, 0]$$

(7) (a) * Global thresholding separates object and background using a threshold value.

* Pixel $T = 1, 0.$

(b) Given

Threshold (200)

since $T = 100$

$$200 > 10$$

$$= [1].$$

(8) Edge Detection

(a) Explain the concept of edge detection

(b) Apply the Sobel operator

$$\begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix} \text{ to } \begin{bmatrix} 10 & 10 & 20 & 20 & 10 \\ 20 & 20 & 20 & 20 & 20 \end{bmatrix}$$

compute the gradient at the center.

(a) * Detects object boundaries

* Used for feature extraction

b) Process: * Apply Sobel kernel
* Add result
Gradient = 80.

9) Region growing

a) group similar pixels

b) process: seed = 50

$$* |48 - 50| = 2 \checkmark$$

$$* |52 - 50| = 2 \checkmark$$

$$* |60 - 50| = 10 \times$$

$$\text{Ans: Region} = 48, 50, 52, 54.$$

10) Smoothing & Thresholding

a) Remove noise before segmentation.

b) Process: $\sigma < 1$ brightness image

$$\text{Cid: } \sqrt{0} = 0, \sqrt{5} = 5$$

$$\sqrt{100} = 10, \sqrt{225} = 15$$

$$\text{Ans: } 0, 5, 10, 15.$$

11) Gamma Transformation.

a) Control Brightness

b) Process: mean = 48, Threshold = 80 $48 < 80 \rightarrow 0$

$$\text{Ans: } 0, 10, 11$$

12) Contrast Stretching

a) improve contrast

(hi, a) process:

$$50 \rightarrow 0$$

$$75 \rightarrow 64$$

$$100 \rightarrow 128$$

$$150 \rightarrow 255$$

$$\text{Ans: } 0, 64, 128, 255$$

14) Histogram Equalization.

- a) Improves image content
- b) CDF preserves intensity order
- c) Process: CDF: 4, 10, 20, 28
- d) Process: New level = $(\text{CDF}(22) \times 255)$
New level = 216, 17, 15
- e) Contrast is improved.

14) Median filtering.

- a) Removes salt & pepper noise
- b) Preserves edges better than averaging.
- c) Process: Arrange value: 20, 24, 25, 26, 27, 28, 29, 30, 35
- d) Median = 27
- e) Noise is removed efficiently.

15) Gaussian smoothing

- a) Reduces image noise
- b) Give more weight to nearby pixels
- c) Center pixel = 27
- d) Smooth pixel with edges.

16) High-pass filtering

- a) Enhance edges
- b) Highlight fine details
- c) Apply high pass mask

17) Frequency Domain High pass

- a) High frequencies contain edge information
- b) Retains smooth background
- c) Remove low frequencies
- d) output: High frequency components
- e) Sharper result.

18) Adaptive Thresholding

- a) Handles lighting
- b) uses local threshold
- c) $\text{threshold} > 100$
- d) $\text{output} \in \{0, 10, 11\}$
- e) Better segmentation.

19) Vertical Sobel

- a) Detects vertical edges
- b) Better noise suppression
- c) Apply sobel kernel
- d) $\text{centre value} = 80$
- e) Strong vertical edges

20) Prewitt edge Detection

- a) Prewitt edge Detection
- b) Simpler than sobel
- c) Apply prewitt kernel
- d) $\text{output} = 80$
- e) Indicated Verts:

21) Region merging

- a) merge similar neighbouring regions
- b) Based on intensity difference
- c) $\text{difference} = 6 \geq 5$
- d) No merging
- e) Region frame is separate.

low-pass filtering

- a) Removes high frequency noise
- b) Prevents low frequencies
- c) Process: Remove high frequencies
- d) Output: low frequency

25) Laplacian Sharpening

- a) Enhances details
- b) Detects edges
- c) Apply Laplacian
- d) Center pixel = 80
- e) Sharper image

26) Histogram matching

- a) matches reference histogram
- b) gives constant appearance
- c) Problem: $10 \rightarrow 20$
 $50 \rightarrow 80$
 $100 \rightarrow 160$
- d) Output: 20, 80, 160
- e) Better brightness

27) Noise Reduction (compression)

- a) salt and pepper noise
- b) Avg & Sum
Median middle
- c) Avg = 210
- d) median = 24
- e) median filter is better

- 28) Combined filtering
- a) Removes noise than sharper
 - b) prevents noise amplification.
 - c) process
Arg: sharper
 - d) final output = SD
 - e) Better image quality.

- 29) Multi thread segmentation
- a) Divides image into multiple region
 - b) user threshold 30 to 100
 - c) process: $20 \times 80 \rightarrow \text{class } 1$
 $60 \rightarrow \text{class } 2$
 $120 \rightarrow \text{class } 3$
 - d) class: 12, 7, 3
 - e) Better object separation.

- 30) Boundary Detection:
- a) Detect object boundary
 - b) user edges.
 - c) process:
Detect edges $\rightarrow$ broken edge
 - d) output: Boundary
 - e) Improves object recognition.